Manufactured solutions for the lead-field ECG

Construction of the exact torso potentials, electrodes and compared quantities

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1 The problem

This document describes the manufactured problems with which cardiacFoam verifies the lead-field ECG: the ECG domain `leadFieldECG`, whose channels are built from the lead fields of the utility `buildLeadFields`. In the method of manufactured solutions (Roache 2002) an exact field is chosen first and the data that make it satisfy the equations are derived from it. The document gives, for each problem, the geometry and its regions, the conductivities, the exact heart transmembrane potential and torso potential, the electrodes, the reference point that fixes the constant of the potential, the meshes, and the quantities the verifier compares. It states the problems only; it contains no solver output. A reader with this document, the generator folder and the tutorial cases can rebuild each problem in another code.

1.1 Domain and governing equations

The conductor is the heart Ω_H (the cellZone `myocardium`) and the torso Ω_T (the bath cellZones of the ECG domain), with the interface $\Gamma = \partial\Omega_H \cap \partial\Omega_T$. For a given transmembrane potential V_m in the heart, the passive potential ϕ solves

$$\begin{cases} \nabla\cdot\left((\mathbf{G}_i+\mathbf{G}_e)\nabla\phi\right) = -\nabla\cdot(\mathbf{G}_i\nabla V_m) + S_H & \text{in } \Omega_H, \\ \nabla\cdot(\sigma\nabla\phi) = S_T & \text{in } \Omega_T, \\ \phi \text{ continuous}, \quad \mathbf{n}\cdot\mathbf{G}_w\nabla\phi = \mathbf{n}\cdot\sigma\nabla\phi & \text{on } \Gamma, \\ \mathbf{n}\cdot\sigma\nabla\phi = g & \text{on } \partial\Omega_T\setminus\Gamma, \end{cases}$$

where S_H, S_T and g are manufactured data (zero in ordinary runs). The heart source is assembled with no wall flux of $\mathbf{G}_i\nabla V_m$. The heart-side wall conductivity $\mathbf{G}_w$ is the conductor's choice: with a monodomain myocardium, whose V_m is sealed ($\mathbf{n}\cdot\mathbf{G}_i\nabla V_m = 0$), the wall carries the total conductivity, $\mathbf{G}_w = \mathbf{G}_i + \mathbf{G}_e$; with a bidomain myocardium the intracellular potential is sealed, $\mathbf{n}\cdot\mathbf{G}_i\nabla(\phi+V_m) = 0$, and $\mathbf{G}_w = \mathbf{G}_e$. The problem is of pure Neumann type; the constant of ϕ is fixed by a reference point $\mathbf{r}_{\text{ref}}$ in the torso (`phiERefPoint` of the torso dictionary), whose cell is pinned to the reference value $\phi_{\text{ref}} = 0$.

The ECG of electrode e at $\mathbf{r}_e$ is the value of ϕ in the cell $c(e)$ that contains $\mathbf{r}_e$. The lead-field ECG computes it from the lead field of the electrode, the solution of the adjoint discrete problem, without solving for ϕ at each sample (Geselowitz 1967); Section 4 states the identity and what is compared. Electrodes lie in the torso: `buildLeadFields` and `leadFieldECG` stop with a fatal error when an electrode cell is a heart cell. Each case has two kinds of electrode, recorded in the same way. The eight body-surface electrodes `surface1` to `surface8` lie just inside the outer boundary of the conductor, in cells with a face on it; they are the ECG. The interior probes lie inside the torso, in the bath ball or next to the heart, and test the same channel where the potential and its error differ from those at the surface.

The equations use SI units with lengths and times normalised by 1 m and 1 s, so the arguments of the trigonometric, exponential and square-root functions are dimensionless.

1.2 Nomenclature

The code name is the key of the cardiacFoam case dictionaries or the name used in the generators and the NumPy reference modules. "Case units" marks a quantity defined only through the manufactured construction, with no physical calibration.

Symbol	Meaning	Unit	Code name
$\mathbf{x} = (x, y, z)$	position, normalised by 1 m	m	`x`, `y`, `z`
t, $S(t)$	time, normalised by 1 s; $\sqrt{1+t}$	s; –	`t`; `St`
Ω_H, Ω_T, Γ	heart (cellZone `myocardium`); torso (the bath cellZones); heart–torso interface	–	`cellZone`; `bathCellZones`
$\mathbf{n}$	unit normal of Γ or of the outer boundary	–	–
V_m	transmembrane potential, the heart input of the conductor	V	`Vm`
ϕ	potential of the passive conductor (heart and torso)	V	`phiE`
$\mathbf{G}_i$, $\mathbf{G}_e$	intracellular and extracellular conductivity tensors	S m^{-1}	`conductivityIntracellular`, `conductivityExtracellular`; `Gi`, `Ge`
σ, σ_B	torso conductivity; its value in the curved conductor	S m^{-1}	`bodyAndOrgansConductivity`; `sigmaB`
k	$G_{i,xx}/(G_{i,xx} + G_{e,xx})$ of the slab	–	`slabK_`; `K` (NumPy)
σ_1, σ_2, x_L, x_R	inner and outer torso layer conductivities; layer interfaces	S m^{-1}; m	`torsoInner`, `torsoOuter`; `layerInterfaces`
α, $R(x)$	surface current of the layered slab; $\int_0^x \mathrm{d}x'/\sigma(x')$	A m^{-2}; m^2 S^{-1}	`surfaceCurrentPatches`; `layered_resistance`
S_H, S_T, g	phiE source in the heart and in the torso; outward current on the outer boundary (curved conductor)	A m^{-3}; A m^{-2}	`phiESource`; `shellSource`
r_1, r_2, r_3	radii of the bath ball, of the heart shell and of the outer sphere	m	`r1`, `r2`, `r3`
λ, ω, θ	transmural coordinate; function vanishing on both spheres; fibre angle	–; m; rad	`lam`, `om` (shell generator)
$\mathbf{f}, \mathbf{s}, \mathbf{e}_c$	fibre, sheet and sheet-normal directions of $\mathbf{G}_i$ and $\mathbf{G}_e$	–	(shell generator)
$\sigma_{il}, \sigma_{it}, \sigma_{in}$; $\sigma_{el}, \sigma_{et}, \sigma_{en}$	eigenvalues of $\mathbf{G}_i$ and of $\mathbf{G}_e$	S m^{-1}	`gil`, `git`, `gin`; `gel`, `get`, `gen`
$\hat{f}$, $\hat{h}$, a_j, b_j	smooth extracellular and intracellular fields and their coefficients	V	`fhat`, `hhat`, `fA`, `hA`
g_j, m_j, d_j, $\mathbf{n}_j$	wall corrections of sphere j; $\partial_n\omega$ on sphere j; normal of sphere j out of the heart	–	`g`, `m` (shell generator)

Symbol	Meaning	Unit	Code name
$\mathbf{r}_e$, $c(e)$, $\mathbf{x}_{c(e)}$	electrode position; the cell that contains it; its centroid	m	`electrodePositions`; `electrodeCells_`; `electrodeCentroids_`
e	an electrode: a body-surface electrode (`surface1` to `surface8`, in a cell with a face on the outer boundary) or an interior probe	–	`electrodePositions`
$\partial\Omega$	outer boundary of the conductor; its cells (a face on it) are the region of `phiStatic_surface`	–	`phiStatic_surface`
$\mathbf{r}_{\text{ref}}$, $\mathbf{x}_{\text{ref}}$, ϕ_{ref}	reference point of the conductor; centroid of its cell; reference value	m; m; V	`phiERefPoint`; `referenceCentroid_`; `phiEReferenceValue`
A, $\mathbf{b}$, $\mathbf{s}$	discrete forward operator; heart source $-\nabla_h \cdot (\mathbf{G}_i \nabla_h V_m)$; extra source of the curved conductor	–	`assemble`; `heartSource`; `shellSource`
$\mathbf{z}_e$, c_e	lead field of electrode e; constant term of its channel	–	`leadField_<e>`; `constants_`
V_c	volume of cell c	m^3	`V()`

1.3 Reading and regenerating

The exact potentials come from two sources. The slab and layered potentials (Section 2) are closed forms in one variable, written in the cardiacFoam header `manufacturedLeadFieldECGReference.H` of `src/verificationModels/ecgVerification/`; this folder holds their NumPy transcription, `docs/leadField/leadfield_reference.py`, which follows the header branch by branch. The curved-conductor potential (Section 3) is the transmission solution of the sympy generator `shell_mms.py` of `tutorials/manufacturedSolutions/generators/`, which writes the C++ header `manufacturedShellTransmissionBathBidomainReference.H` of `src/verificationModels/bathBidomainVerification/`.

From the generator folder, with the environment of `requirements.txt`:

```
python docs/leadField/leadfield_reference.py --self-check
python docs/leadField/leadfield_reference.py --check-run RUN
python docs/leadField/figures.py [--mesh-case NAME=PATH ...]
quarto render docs/leadField/leadField_mms.qmd --to pdf
```

`--check-run RUN` reads a finished run of a lead-field tutorial (its `log.cardiacFoam`, `postProcessing/manufacturedLeadFieldECG_ECG.dat` and `constant/polyMesh`) and compares the exact columns written by the verifier with the NumPy module (Section 2.5, Section 3.5).

2 Insulated slab and layered torso

2.1 Description and limitations

A heart slab between two torso slabs, with a one-dimensional exact solution along x. The tutorial is `manufacturedSolutions/leadFieldECG`, with two exact solutions selected by `-problem`:

- `slab`: every outer face is insulated. The torso potential is constant in each torso slab, so the ECG between the slabs is known exactly and does not depend on the torso conductivity. These are the fields of the FDA bath slab of the bath-bidomain tutorial with the bath current set to zero.
- `layered`: each torso slab is split into two layers of different conductivity, and a uniform current density crosses the whole box along x. The potential is then piecewise linear in the torso, with a change of slope at each layer interface and at each heart wall.

The heart runs the one-dimensional FDA monodomain or bidomain manufactured field (Pathmanathan and Gray 2014) on the three-dimensional heart mesh (`dimension "1D"`), so the computed V_m approximates the exact V_m below and the lead-field and direct channels can be compared on it.

The problems exercise the heart source $-\nabla \cdot (\mathbf{G}_i \nabla V_m)$, the heart and torso operators, two plane heart–torso interfaces and, in the layered problem, two plane torso–torso interfaces with a conductivity ratio of 5 and a current through all of them. They do not exercise a curved interface, a varying torso conductivity, an off-diagonal conductivity or an electrode where the exact potential varies across the slab's y and z. In the slab problem the exact torso potential is constant, so the point and the centroid values of an electrode are equal and the torso conductivity does not enter. The layered problem with a current needs the monodomain myocardium: at the walls the exact heart slope $\alpha/(\mathbf{G}_i + \mathbf{G}_e)_{xx}$ carries the intracellular flux $\alpha G_{i,xx}/(G_{i,xx} + G_{e,xx})$, which the sealed intracellular potential of the bidomain conductor does not admit.

2.2 Problem instance

Box $[-1, 2] \times [0, 1]^2$: torso $-1 \le x \le 0$, heart $0 \le x \le 1$, torso $1 \le x \le 2$ (Figure 1). The heart conductivities are uniform and diagonal (S m^{-1}):

myocardium	conductivity	ECG domain $\mathbf{G}_i$	ECG domain $\mathbf{G}_e$
`monodomain`	$\mathbf{G}_m =$ diag(0.111453302, 0.121585420, 0.030396355)	$(2+\sqrt{2})\,\mathbf{G}_m$	$\sqrt{2}\,\mathbf{G}_m$
`bidomain`	$\mathbf{G}_i =$ diag(0.111453302, 0.121585420, 0.030396355), $\mathbf{G}_e =$ diag(0.046165469, 0.050362330, 0.012590582)	the solver's	the solver's

In both rows $k = G_{i,xx}/(G_{i,xx} + G_{e,xx}) = 1/\sqrt{2}$. The exact solution depends on the xx entries only, so it is the same for the options `-conductivity anisotropic`, `isotropic` and `unequal` of the tutorial; the three differ in what the static error measures (tutorial README).

- `slab`: torso conductivity $\sigma = 0.0230827345$ in the cellZone `bath`; every outer face insulated ($g = 0$); $S_H = S_T = 0$.

- `layered`: the torso slabs split at $x_L = -0.5$ and $x_R = 1.5$ into the cellZones `torsoInner` ($x_L < x < 0$ and $1 < x < x_R$, $\sigma_1 = 0.1$) and `torsoOuter` ($x < x_L$ and $x > x_R$, $\sigma_2 = 0.02$); surface currents $\mathbf{n} \cdot \sigma \nabla \phi = -\alpha$ on the face `xMin` ($x = -1$) and $+\alpha$ on `xMax` ($x = 2$), $\alpha = 0.01$ (`-alpha`); the faces `sides` insulated. The current density $-\sigma \nabla \phi = -\alpha\, \mathbf{e}_x$ is the same in every layer.
- Reference point $\mathbf{r}_{\text{ref}} = (-0.9013, 0.0437, 0.0437)$ in the left torso (the outer layer in `layered`), $\phi_{\text{ref}} = 0$. The bidomain myocardium also pins its own heart potential at (0.1137, 0.0437, 0.0437); that pin belongs to the heart solve and not to the conductor of the ECG domain.
- Electrodes: eight on the body surface, 0.003 inside the outer faces of the torso slabs (two on each end face $x = -1$, $x = 2$ and one on each of the four side faces), and five interior probes. None lies on a hexahedral face at $N = 10, 20, 40, 80$.

electrode	$\mathbf{r}_e$	region (`slab`; `layered`)	exact ϕ, `slab`
`surface1`	(−0.997, 0.3137, 0.6871)	face $x = -1$; `torsoOuter` (pinned layer)	0
`surface2`	(−0.997, 0.7213, 0.2419)	face $x = -1$; `torsoOuter` (pinned layer)	0
`surface3`	(1.997, 0.4127, 0.5813)	face $x = 2$; `torsoOuter`	$2kS$
`surface4`	(1.997, 0.8317, 0.1733)	face $x = 2$; `torsoOuter`	$2kS$
`surface5`	(−0.3713, 0.003, 0.4419)	face $y = 0$, left torso; `torsoInner`	0
`surface6`	(−0.6871, 0.5371, 0.997)	face $z = 1$, left torso; `torsoOuter` (pinned layer)	0
`surface7`	(1.3317, 0.997, 0.6137)	face $y = 1$, right torso; `torsoInner`	$2kS$
`surface8`	(1.7219, 0.2713, 0.003)	face $z = 0$, right torso; `torsoOuter`	$2kS$
`torsoLeftFar`	(−0.8713, 0.2311, 0.7129)	left torso; `torsoOuter` (pinned layer)	0
`torsoLeft`	(−0.4567, 0.4137, 0.5871)	left torso; `torsoInner`	0
`nearLeft`	(−0.0213, 0.4521, 0.5317)	left torso next to the heart; `torsoInner`	0
`nearRight`	(1.0187, 0.3719, 0.4613)	right torso next to the heart; `torsoInner`	$2kS$
`torsoRight`	(1.5433, 0.6173, 0.3311)	right torso; `torsoOuter`	$2kS$

- Time: $\Delta t = 0.01$, end time 0.05; the verifier samples every step.

In cardiacFoam:

- tutorial `manufacturedSolutions/leadFieldECG`, options `-problem slab|layered`, `-alpha`, `-mesh hex|tet`, `-N`, `-myocardium monodomain|bidomain`;
- ECG domain `leadFieldECG` with the `torso` sub-dictionary: the `bathCellZones`, the field `bodyAndOrgansConductivity`, the interface interpolation `distanceWeightedHarmonic`, the reference point, and for `layered` the `surfaceCurrentPatches`;
- verifier `manufacturedLeadFieldECGVerifier` with `reference slab`, or `reference layered` and `layerInterfaces (-0.5 1.5)`.

2.3 Construction

Slab. With $S = \sqrt{1+t}$, the heart field is the FDA field of $d = 1$, $V_m = S\cos(\pi x)$. Its derivative vanishes at $x = 0$ and $x = 1$, so no current leaves the heart through Γ whichever wall conductivity the conductor uses. In the heart, $\phi = -kV_m + C$ solves $\nabla \cdot ((\mathbf{G}_i + \mathbf{G}_e)\nabla\phi) = -\nabla \cdot (\mathbf{G}_i \nabla V_m)$ because ϕ and V_m depend on x only and $(G_{i,xx} + G_{e,xx})\, k = G_{i,xx}$. Its slope vanishes at both walls, so the torso potential is constant on each side and continuity fixes it to the wall value: $-kS + C$ on the left and $kS + C$ on the right. The torso conductivity does not enter.

Layered. A uniform current density $-\alpha\, \mathbf{e}_x$ through layers of conductivity $\sigma(x)$ (σ_2, σ_1, $(G_i + G_e)_{xx}$, σ_1, σ_2 from left to right) has the potential $\alpha R(x)$ with $R' = 1/\sigma$. R is continuous and $\sigma R' = 1$ in every layer, so the potential and the normal current are continuous at $x = x_L, 0, 1, x_R$; on `xMin` the outward normal is $-\mathbf{e}_x$ and $\mathbf{n} \cdot \sigma\nabla(\alpha R) = -\alpha$, on `xMax` it is $+\alpha$. Adding αR to the slab potential leaves the heart equation unchanged, because R is linear in the heart, and V_m is the slab's. The sides stay insulated because the potential depends on x only and no tensor has xy or xz entries.

Gauge. The verifier evaluates the exact potential at a point $\mathbf{x}$ as

$$\phi(\mathbf{x}, t) = \phi^*(\mathbf{x}, t) - \phi^*(\mathbf{x}_{\mathrm{ref}}, t) + \phi_{\mathrm{ref}},$$

with ϕ^* the formula below and $\mathbf{x}_{\mathrm{ref}}$ the centroid of the cell that contains $\mathbf{r}_{\mathrm{ref}}$, the cell the conductor pins. The constant C of the formulas therefore cancels.

2.4 Exact solution

$$V_m = S\cos(\pi x) \quad (0 \le x \le 1), \qquad S = \sqrt{1+t}, \qquad k = \frac{G_{i,xx}}{G_{i,xx} + G_{e,xx}},$$

$$\phi^*_{\mathrm{slab}}(x,t) = \begin{cases} -kS + C, & x \le 0, \\ -kS\cos(\pi x) + C, & 0 < x \le 1, \\ kS + C, & x > 1, \end{cases} \qquad C = kS,$$

so that, gauged in the left torso, $\phi = 0$ for $x \le 0$, $\phi = kS(1 - \cos\pi x)$ in the heart and $\phi = 2kS$ for $x \ge 1$: the exact ECG between the right and the left torso is $2kS$. In the layered problem

$$\phi^*_{\mathrm{layered}}(x,t) = \phi^*_{\mathrm{slab}}(x,t) + \alpha R(x),$$

$$R(x) = \begin{cases} \dfrac{x_L}{\sigma_1} + \dfrac{x - x_L}{\sigma_2}, & x \le x_L, \\ \dfrac{x}{\sigma_1}, & x_L < x \le 0, \\ \dfrac{x}{(G_i + G_e)_{xx}}, & 0 < x \le 1, \\ \dfrac{1}{(G_i + G_e)_{xx}} + \dfrac{x - 1}{\sigma_1}, & 1 < x \le x_R, \\ \dfrac{1}{(G_i + G_e)_{xx}} + \dfrac{x_R - 1}{\sigma_1} + \dfrac{x - x_R}{\sigma_2}, & x > x_R. \end{cases}$$

The verifier takes α from the surface currents and the four layer conductivities from the torso cells, and stops with a fatal error when a torso cell crosses a layer interface, when a layer has more than one or an anisotropic conductivity, when the surface currents are not along x or carry a net

current, or when a current is combined with the bidomain conductor. With $\alpha = 0$ the layered potential equals the slab's for any σ_1, σ_2.

In code, the functions of `manufacturedLeadFieldECGReference.H`:

- `manufacturedLeadFieldSlabVm`;
- `manufacturedLeadFieldSlabPhi`, which calls `manufacturedFDABathPhiEAt` with the bath current 0 and $\sigma_e = 1$;
- `manufacturedLeadFieldLayeredResistance`;
- `manufacturedLeadFieldLayeredPhi`;

and in NumPy `slab_vm`, `slab_phi_raw`, `layered_resistance`, `layered_phi_raw` and `gauged` of `leadfield_reference.py`.

2.5 Self-checks

`leadfield_reference.py --self-check` asserts that $k = 1/\sqrt{2}$ to 10^{-11}; that ϕ^*_{slab} is 0 on the left, $2kS$ on the right and continuous at $x = 0$ and $x = 1$; that R is continuous with $\sigma R' = 1$ (central differences, 10^{-8}) on both sides of each of x_L, 0, 1, x_R; and that the electrodes of the case are `surface1` to `surface8`, 0.003 from the outer boundary, and probes farther inside, all in the torso and none on a hexahedral face at $N = 10, 20, 40, 80$.

`--check-run RUN` reads the verifier line of `log.cardiacFoam` (reference, k, α and the layer conductivities, which must equal the values of Section 2 to the printed precision) and, at every sample, asserts to 10^{-12} that:

- $\phi^*(\mathbf{r}_e, t) - \phi_{\text{exactPoint},e}$ is one constant over the electrodes, the gauge;
- the gauge equals $\phi^*(\mathbf{x}_{\text{ref}}, t)$, with $\mathbf{x}_{\text{ref}}$ the centroid of the cell that contains $\mathbf{r}_{\text{ref}}$;
- $\phi^*(\mathbf{x}_{c(e)}, t)$ minus the gauge equals the column `exactCentroid` of each electrode.

The centroids are computed from the cell vertices in double precision; this is exact for the axis-aligned hexahedra and for tetrahedra. On the meshes of the figures below the largest differences are 3×10^{-15} (`slab`, hex), 6×10^{-14} (`slab`, tet), 3×10^{-15} (`layered`, hex) and 6×10^{-14} (`layered`, tet).

2.6 Figures

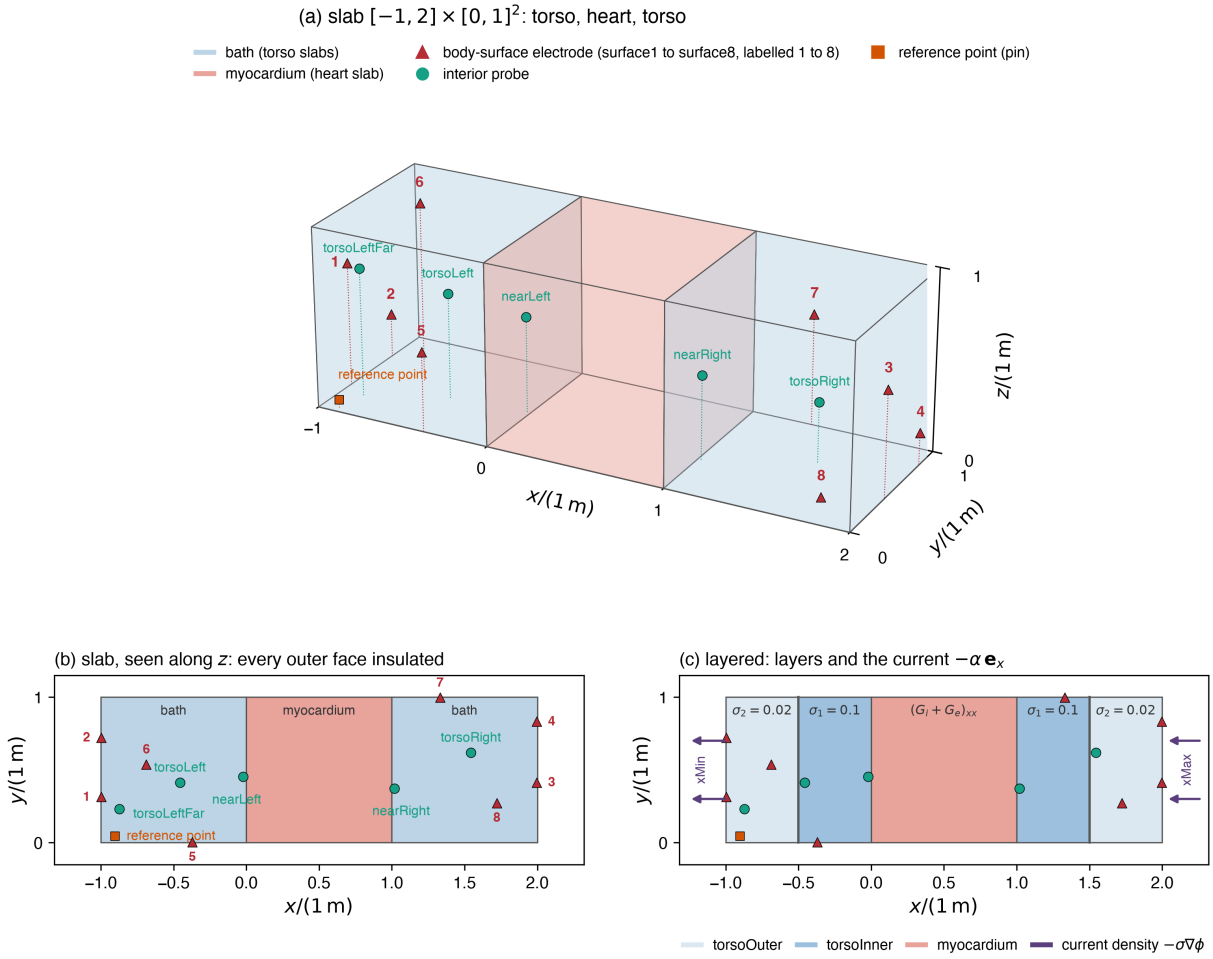


Figure 1: Slab geometry. (a) The box with the torso slabs (cellZone `bath`), the heart slab (`myocardium`), the eight body-surface electrodes (triangles, labelled 1 to 8 for `surface1` to `surface8`), the five interior probes (circles, labelled by name) and the reference point (square); dotted lines drop each point to $z = 0$. (b) The same, seen along z. (c) The layered problem: the layers `torsoOuter` and `torsoInner` with their conductivities, the heart, the electrodes and probes, and the current density $-\sigma\nabla\phi = -\alpha\,\mathbf{e}_x$, which enters through `xMax` and leaves through `xMin`.

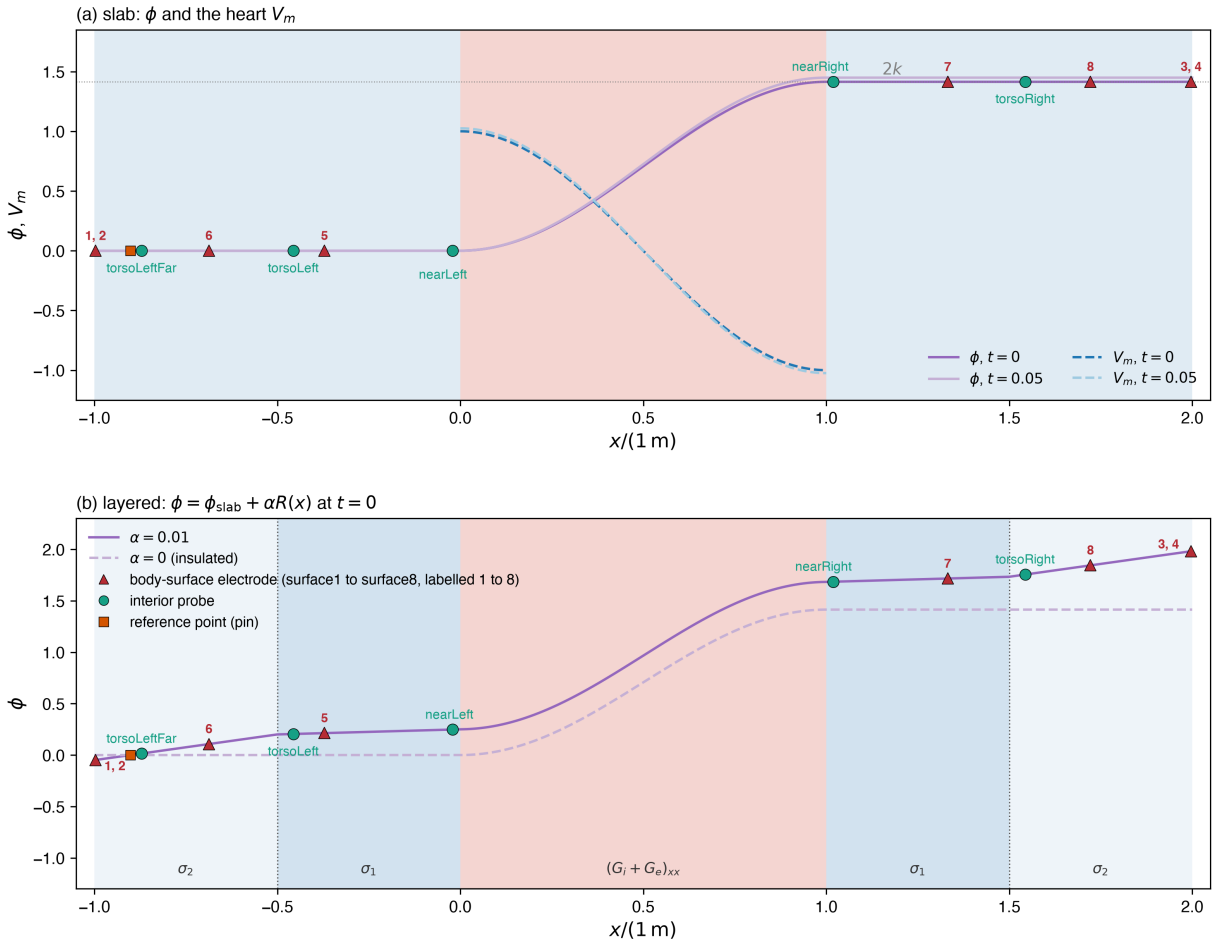
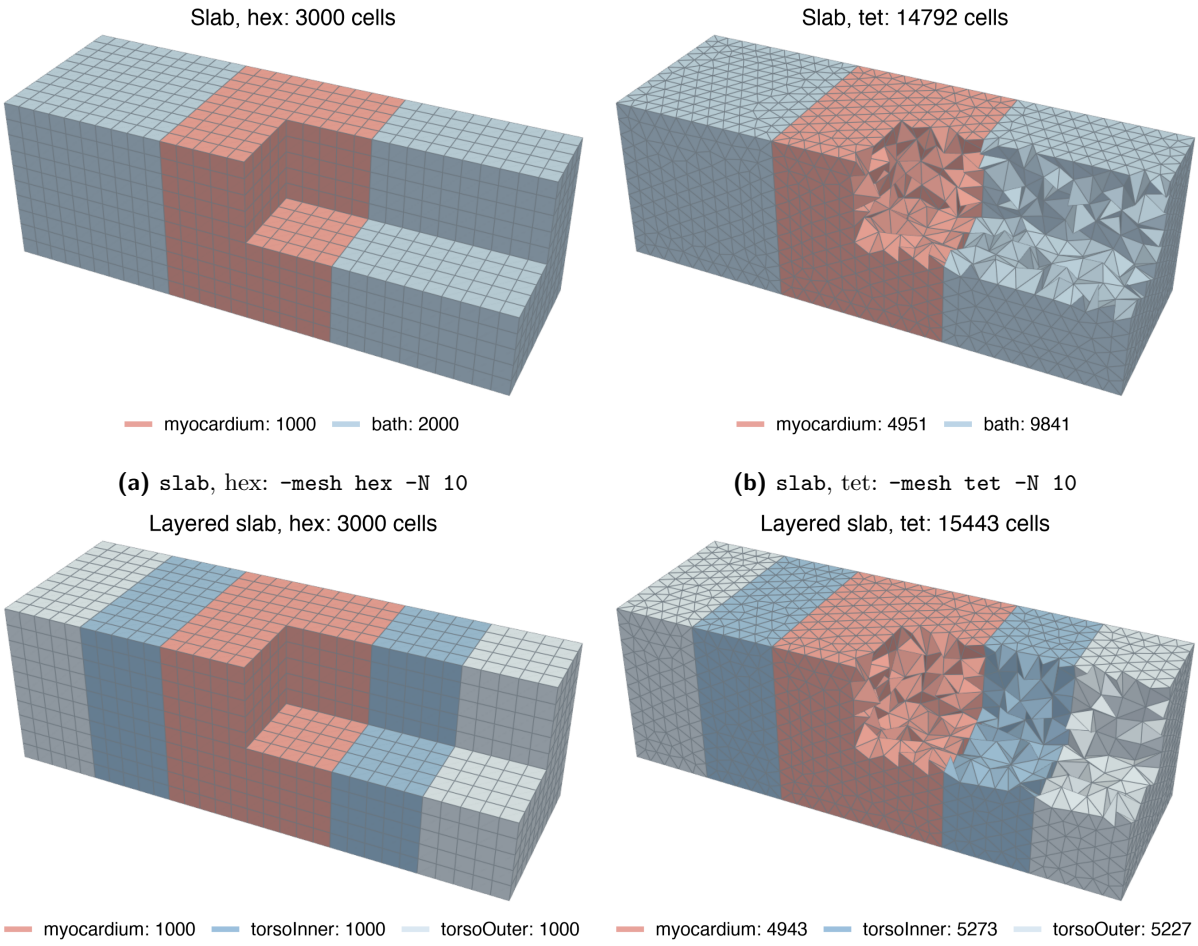


Figure 2: Exact potentials along x (they do not depend on y, z), gauged to 0 at the reference point $\mathbf{r}_{\text{ref}}$ (square); the verifier gauges at the centroid of its cell, which shifts every curve by the same constant. Triangles mark the body-surface electrodes at their x (labelled by number; `surface1` and `surface2`, and `surface3` and `surface4`, share an x and a label), circles the interior probes. (a) Slab: ϕ and the heart V_m at $t = 0$ and $t = 0.05$; ϕ is constant in each torso slab. (b) Layered, $t = 0$: $\phi = \phi_{\text{slab}} + \alpha R$ with $\alpha = 0.01$ and with $\alpha = 0$; the slope changes at the layer interfaces (dotted) and at the heart walls.



(a) `slab`, hex: `-mesh hex -N 10` **(b)** `slab`, tet: `-mesh tet -N 10`

(c) `layered`, hex: `-problem layered -mesh hex -N 10` **(d)** `layered`, tet: `-problem layered -mesh tet -N 10`

Figure 3: Meshes of the tutorial `leadFieldECG` at its default $N = 10$, coloured by cellZone, made by its `Allrun`: (a) `blockMesh`, three blocks of N^3 cells; (b) `gmsh` Delaunay tetrahedra of size $1/N$, seed 1, with the planes $x = 0$ and $x = 1$ as shared internal faces, converted by `gmshToFoam`; (c) `blockMesh`, five blocks with $N/2$ cells along x in each torso layer, so the layer interfaces are mesh faces; (d) as (b) with the planes $x = x_L$, $x = x_R$ also shared. `topoSet` builds the cellZones. The octant facing the viewer is removed cell by cell.

2.7 Regeneration

```
python docs/leadField/leadfield_reference.py --self-check
python docs/leadField/leadfield_reference.py --check-run SLAB_RUN \
    --check-run LAYERED_RUN
python docs/leadField/figures.py --mesh-case slab_hex=SLAB_HEX \
    --mesh-case slab_tet=SLAB_TET --mesh-case layered_hex=LAYERED_HEX \
    --mesh-case layered_tet=LAYERED_TET
```

A run is a copy of the tutorial outside the repository after `./Allrun [-problem layered] -mesh hex|tet -N 10`. The C++ reference is hand-written; there is no generator to compare it with, and the run check above is the comparison.

3 Curved conductor

3.1 Description and limitations

A heart shell $r_1 < |\mathbf{x}| < r_2$ between a bath ball $|\mathbf{x}| < r_1$ (cellZone `bloodPool`) and an outer bath $r_2 < |\mathbf{x}| < r_3$ (cellZone `bath`), with the transmission solution of `shell_mms.py`. The current generated in the heart crosses both curved interfaces into the torso. The heart conductivities are full tensors whose fibre and sheet rotate through the wall, and the bath conductivity varies in space. The exact potential has non-zero normal and tangential gradients on both spheres; it is continuous across them, the extracellular flux is continuous and the intracellular potential is sealed, all exactly.

The problem exercises the heart source with a full, varying $\mathbf{G}_i$, the anisotropic heart operator, two curved heart–torso interfaces crossed by polyhedral faces, a varying torso conductivity, and the bidomain wall condition. It also needs data that an ordinary run does not have: a source in every zone and a current through the outer boundary. These enter both channels as one extra cell source (Section 4), so the problem verifies the conductor and the lead fields with the manufactured data included. The verifier uses the exact V_m at the heart cell centres only; the myocardium of the case runs one step of `BuenoOrovio`, whose V_m is uniform, and the channel on the computed V_m is a check of reciprocity, not of accuracy. The solution needs the bidomain conductor and the reference point; ground and surface-current patches are refused.

3.2 Problem instance

parameter	value	parameter	value
$r_1,\ r_2,\ r_3$	0.5, 1, 1.5	$\gamma_1,\ \gamma_2$	20°, 50°
$\sigma_{il},\ \sigma_{it},\ \sigma_{in}$	0.17, 0.019, 0.01	$\theta_0,\ \theta_1$	−20°, 90°
$\sigma_{el},\ \sigma_{et},\ \sigma_{en}$	0.62, 0.12, 0.08	k	π
$\sigma_{0,\mathrm{in}},\ \sigma_{0,\mathrm{out}},\ \varepsilon$	0.6, 0.25, 0.2	$a_0, \ldots, a_3$	0.3, 0.2, 0.4, 0.15
A_{cm}	0	$b_0,\ b_1,\ b_2$	−0.1, 0.3, 0.25

These are `default_params()` of `shell_mms.py` and `defaultParams()` of the header. The outer sphere r_3 is the extent of the mesh; the formulas hold everywhere. Conductivities are in S m^{-1}, angles in degrees.

- Reference point $\mathbf{r}_{\mathrm{ref}} = (1.3137, 0.0371, 0.0459)$ in the outer bath, $\phi_{\mathrm{ref}} = 0$. The bidomain heart solve pins its own potential at (0.7413, 0.0317, 0.0291), in the heart; that pin does not enter the conductor.
- Electrodes: eight on the body surface, spread over the outer sphere at $|\mathbf{x}| = 1.495$, inside its faceted surface at every level and off the coordinate planes and the block seams of the cubed sphere, and four interior probes:

electrode	$\mathbf{r}_e$	$\lvert\mathbf{r}_e\rvert$	zone
`surface1`	(1.2436, 0.8151, −0.1553)	1.495	`bath`, body surface
`surface2`	(0.2050, 0.6740, 1.3186)	1.495	`bath`, body surface
`surface3`	(1.1945, −0.8631, −0.2517)	1.495	`bath`, body surface
`surface4`	(−0.2551, 1.4080, −0.4329)	1.495	`bath`, body surface
`surface5`	(−0.3258, −0.8777, 1.1656)	1.495	`bath`, body surface

electrode	$\mathbf{r}_e$	$\lvert\mathbf{r}_e\rvert$	zone
`surface6`	$(0.1488, -0.3633, -1.4425)$	1.495	`bath`, body surface
`surface7`	$(-1.4160, 0.4317, 0.2089)$	1.495	`bath`, body surface
`surface8`	$(-0.8375, -1.1858, -0.3569)$	1.495	`bath`, body surface
`outerNear`	$(0.0517, 1.0713, 0.0811)$	1.076	`bath`, next to the heart
`outerSide`	$(-1.2117, -0.4313, 0.6271)$	1.431	`bath`
`outerFar`	$(0.9113, 0.8121, -0.6730)$	1.394	`bath`
`pool`	$(0.1131, -0.0733, 0.2117)$	0.251	`bloodPool`

- One time step, $\Delta t = 10^{-3}$; at that time $S = \sqrt{1.001}$.

In cardiacFoam:

- tutorial `manufacturedSolutions/leadFieldECG/sphericalShell`, options `-mesh hex|tet`, `-N`, `-seed`;
- `setManufacturedFields` writes `0/ConductivityIntracellular`, `0/ConductivityExtracellular` and `0/bodyAndOrgansConductivity` from the solution at the cell centres; every cell takes $\mathbf{G}_i$ and $\mathbf{G}_e$ of the heart formula, and the torso cells σ_B of their zone;
- ECG domain `leadFieldECG`, torso cellZones `bloodPool` and `bath`, `distanceWeightedHarmonic`; myocardium `bidomainSolver` with `sealedHeartBoundary true` and `conductivitySource field`;
- verifier `manufacturedLeadFieldECGVerifier`, `reference sphericalShell`.

3.3 Construction

Transmural coordinate and tensors. As in the sealed monodomain shell, with $\rho = |\mathbf{x}|^2$,

$$\lambda = \frac{\rho - r_1^2}{r_2^2 - r_1^2}, \qquad \theta = \theta_0 + \theta_1 \lambda, \qquad \omega = \frac{(\rho - r_1^2)(\rho - r_2^2)}{(r_2^2 - r_1^2)(r_1 + r_2)},$$

$$[\mathbf{e}_a\ \mathbf{e}_b\ \mathbf{e}_c] = R_z(\gamma_1) R_x(\gamma_2), \qquad \mathbf{f} = \cos\theta\, \mathbf{e}_a + \sin\theta\, \mathbf{e}_b, \qquad \mathbf{s} = -\sin\theta\, \mathbf{e}_a + \cos\theta\, \mathbf{e}_b,$$

$$\mathbf{G}_i = \sigma_{il}\, \mathbf{f} \otimes \mathbf{f} + \sigma_{it}\, \mathbf{s} \otimes \mathbf{s} + \sigma_{in}\, \mathbf{e}_c \otimes \mathbf{e}_c, \qquad \mathbf{G}_e = \sigma_{el}\, \mathbf{f} \otimes \mathbf{f} + \sigma_{et}\, \mathbf{s} \otimes \mathbf{s} + \sigma_{en}\, \mathbf{e}_c \otimes \mathbf{e}_c.$$

$\mathbf{G}_i$ and $\mathbf{G}_e$ share one frame and one fibre angle. A fixed frame is used because a fibre field tangent to a sphere vanishes somewhere on it.

Bath conductivity.

$$\hat{\sigma} = \sin\!\left(kx + \tfrac{1}{5}\right) \cos(ky) \left(1 + \tfrac{1}{2} \sin kz\right), \qquad \sigma_B = \begin{cases} \sigma_{0,\mathrm{in}}(1 + \varepsilon\hat{\sigma}), & |\mathbf{x}| < r_1, \\ \sigma_{0,\mathrm{out}}(1 + \varepsilon\hat{\sigma}), & |\mathbf{x}| > r_2. \end{cases}$$

Smooth fields. With $S = \sqrt{1 + t}$, $f = S\hat{f}$ and $h = S\hat{h}$,

$$\hat{f} = a_0 x + a_1 yz + a_2 \sin(kx) \left(1 + \tfrac{1}{2} y\right) + a_3 \cos\!\left(ky + \tfrac{3}{10}\right) e^{2z/5},$$

$$\hat{h} = b_0 z + b_1 \cos(ky) \left(1 - \tfrac{3}{10} x^2\right) + b_2 x \sin(kz).$$

Interface corrections. On sphere j ($j = 1$: $|\mathbf{x}| = r_1$, $j = 2$: $|\mathbf{x}| = r_2$) the unit normal out of the heart is $\mathbf{n}_1 = -\mathbf{x}/r_1$, $\mathbf{n}_2 = \mathbf{x}/r_2$, and $\nabla\omega = d_j\mathbf{n}_j$ with $d_1 = 2r_1/(r_1 + r_2)$, $d_2 = 2r_2/(r_1 + r_2)$. With $\sigma_{B,1}$ the bath-ball formula and $\sigma_{B,2}$ the outer-bath formula of σ_B, evaluated at $\mathbf{x}$,

$$g_j = \frac{\mathbf{n}_j \cdot (\sigma_{B,j}\mathbf{I} - \mathbf{G}_e)\nabla f}{(\mathbf{n}_j \cdot \mathbf{G}_e\mathbf{n}_j)\, d_j}, \qquad m_j = -\frac{\mathbf{n}_j \cdot \mathbf{G}_i\nabla h}{(\mathbf{n}_j \cdot \mathbf{G}_i\mathbf{n}_j)\, d_j},$$

$$g = (1 - \lambda)g_1 + \lambda g_2, \qquad m = (1 - \lambda)m_1 + \lambda m_2.$$

The heart extracellular and intracellular potentials are $\phi = f + \omega g$ and $\phi_i = h + \omega m$, and the bath potential is $\phi = f$. On sphere j, $\omega = 0$ and $\nabla(\omega g) = g_j d_j \mathbf{n}_j$, so

- ϕ is continuous (f on both sides);
- $\mathbf{n}_j \cdot \mathbf{G}_e\nabla\phi = \mathbf{n}_j \cdot \mathbf{G}_e\nabla f + \mathbf{n}_j \cdot (\sigma_{B,j}\mathbf{I} - \mathbf{G}_e)\nabla f = \mathbf{n}_j \cdot \sigma_B\nabla\phi$;
- $\mathbf{n}_j \cdot \mathbf{G}_i\nabla\phi_i = \mathbf{n}_j \cdot \mathbf{G}_i\nabla h - \mathbf{n}_j \cdot \mathbf{G}_i\nabla h = 0$.

The denominators are positive because the tensors are positive definite, so every zone formula extends smoothly beyond its sphere, where polyhedral faces put cell centres.

Common mode. The generator also adds $T(t)$ times a common term to ϕ and ϕ_i (leaving V_m unchanged), with $T(t) = A_{\mathrm{cm}} \sin(\Omega_{\mathrm{cm}} t)$. The case uses $A_{\mathrm{cm}} = 0$, so the term vanishes; the alternate parameters of the generator switch it on for the header self-test.

Data. The sources are the residuals of the exact fields,

$$S_H = \nabla \cdot (\mathbf{G}_i\nabla\phi_i) + \nabla \cdot (\mathbf{G}_e\nabla\phi) \ \text{ in } \Omega_H, \qquad S_T = \nabla \cdot (\sigma_B\nabla\phi) \ \text{ in } \Omega_T,$$

which is the heart equation of Section 1 with $V_m = \phi_i - \phi$, and the outer boundary carries the exact outward current $g_{\mathrm{out}} = \sigma_B\, \mathbf{n} \cdot \nabla\phi$ evaluated at each outer face centre with the outer-bath formula.

3.4 Exact solution

$$\phi = \begin{cases} S\hat{f}, & |\mathbf{x}| < r_1 \text{ and } |\mathbf{x}| > r_2, \\ S\hat{f} + \omega g, & r_1 \le |\mathbf{x}| \le r_2, \end{cases} \qquad V_m = S(\hat{h} - \hat{f}) + \omega(m - g) \quad \text{in } \Omega_H,$$

with g and m linear in S. The verifier evaluates ϕ with the formula of the zone that contains the point (`zoneOf`), gauged as in Section 2 at the centroid $\mathbf{x}_{\mathrm{ref}}$ of the pinned cell. In code: `Vm`, `phiE`, `phiI`, `gradPhiE`, `Gi`, `Ge`, `sigmaB`, `divGiGradPhiI` and `phiESource` (the source S_H or S_T of the zone) of the namespace `shellTransmissionMMS`; in Python `Model.eval_xyz(zone, name, points, t)` of `shell_mms.py`, and `shell_eval`, `shell_phi` of `leadfield_reference.py`.

3.5 Self-checks

`shell_mms.py` runs, before it writes the header, and `test_shell_mms.py` repeats:

- interface conditions: continuity of ϕ, of the extracellular flux and the sealing of ϕ_i below 10^{-12} at 24 points of each sphere and three times, in 30-digit arithmetic, for the default and the alternate parameters;
- sources and gradients: against fourth-order central differences ($h = 10^{-3}$) to 10^{-6} relative at 40 random points, a point next to the origin and the origin, at two times, and with each zone formula evaluated 0.025 beyond its sphere;
- boundedness and positive denominators over the zones widened by 0.05;
- richness: on each sphere and for each tensor, the tangential part of $\mathbf{Gn}$ applied to the gradient ($\nabla\phi$ for $\mathbf{G}_e$, $\nabla\phi_i$ for $\mathbf{G}_i$) has an RMS of at least 0.2 times that of the reference flux ($\mathbf{n}\cdot\mathbf{G}_e\nabla\phi$, $|\mathbf{G}_i\nabla\phi_i|$), and reaches 0.2 of it over at least 70% of the area;
- the common mode leaves V_m, ∇V_m unchanged (10^{-14}), and the manufactured ionic closure of the heart holds (10^{-13}).

The verifier stops when a cell centre lies farther than its own size outside the radial range of its cellZone, when the case conductivities differ from the solution's at the cell centres by more than 10^{-10} relative, or when an electrode lies in the heart. `leadfield_reference.py --self-check` asserts that the twelve electrodes and the reference point lie in the zones of the table above and that the body-surface electrodes lie at $|\mathbf{x}| = 1.495$, and `figures.py` that $\mathbf{G}_i$ and $\mathbf{G}_e$ have the same eigenvectors. `--check-run` compares the run with `shell_mms.py` as in Section 2.5. On the tetrahedra of the figures the gauge and the centroid values agree to 3×10^{-16}. On the cubed sphere the cells are not affine and their outer faces are curved: the pinned-cell centroid is not recomputed, and the centroid values are checked at the six-digit centroids of the log, to 3×10^{-6} at $N = 8$ and $N = 12$.

3.6 Figures

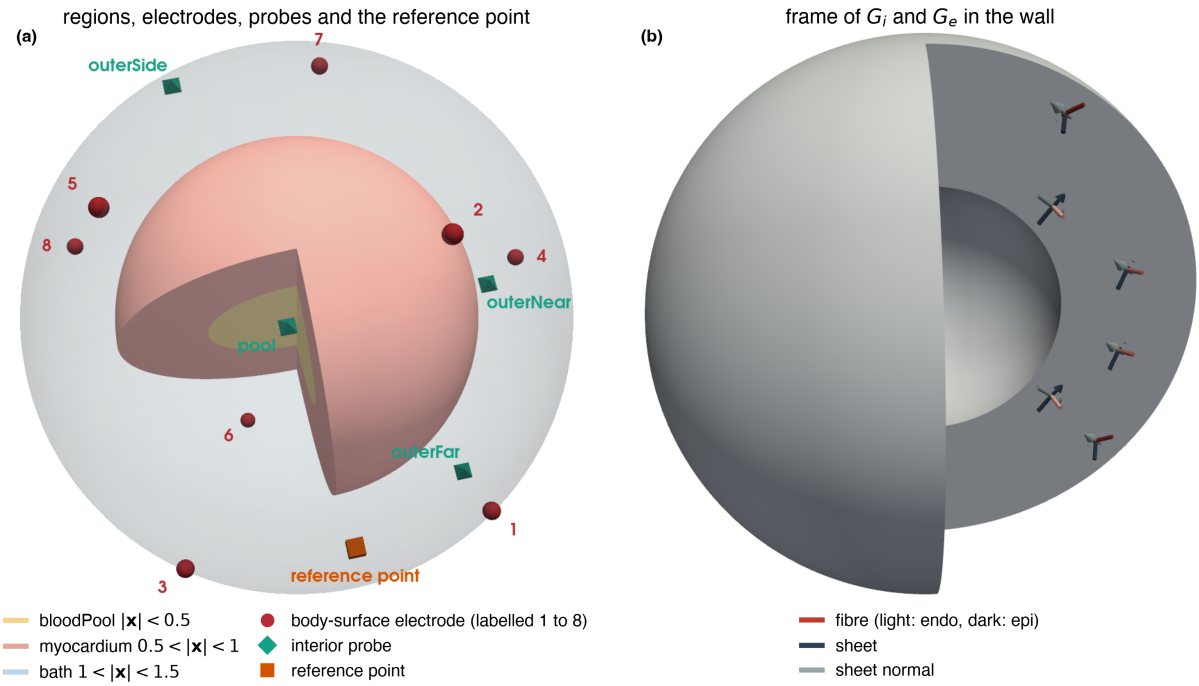


Figure 4: Curved conductor. (a) Cut-away of the bath ball `bloodPool` and the heart shell `myocardium` with the quarter $-90° < \arctan(y/x) < 0$ removed, inside the whole outer bath `bath` (translucent); the eight body-surface electrodes on the outer sphere (spheres, labelled 1 to 8 for `surface1` to `surface8`), the four interior probes (octahedra, labelled by name) and the reference point (cube). `pool` lies in the bath ball, inside the removed quarter. The view direction is chosen so that no opaque zone hides any part of a marker, and each label is placed in the direction farthest from the other markers and labels. (b) The fibre, sheet and sheet-normal directions of $\mathbf{G}_i$ and $\mathbf{G}_e$ (one frame) at six sites through the wall on a cut face, the fibre shaded from the inner (light) to the outer (dark) sphere.

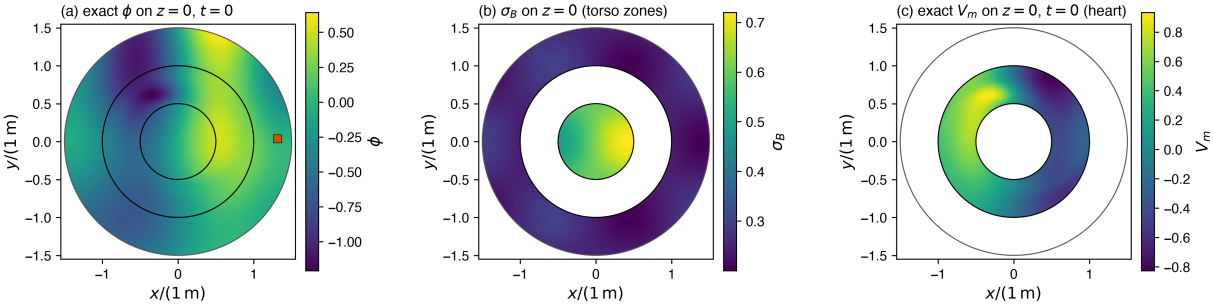
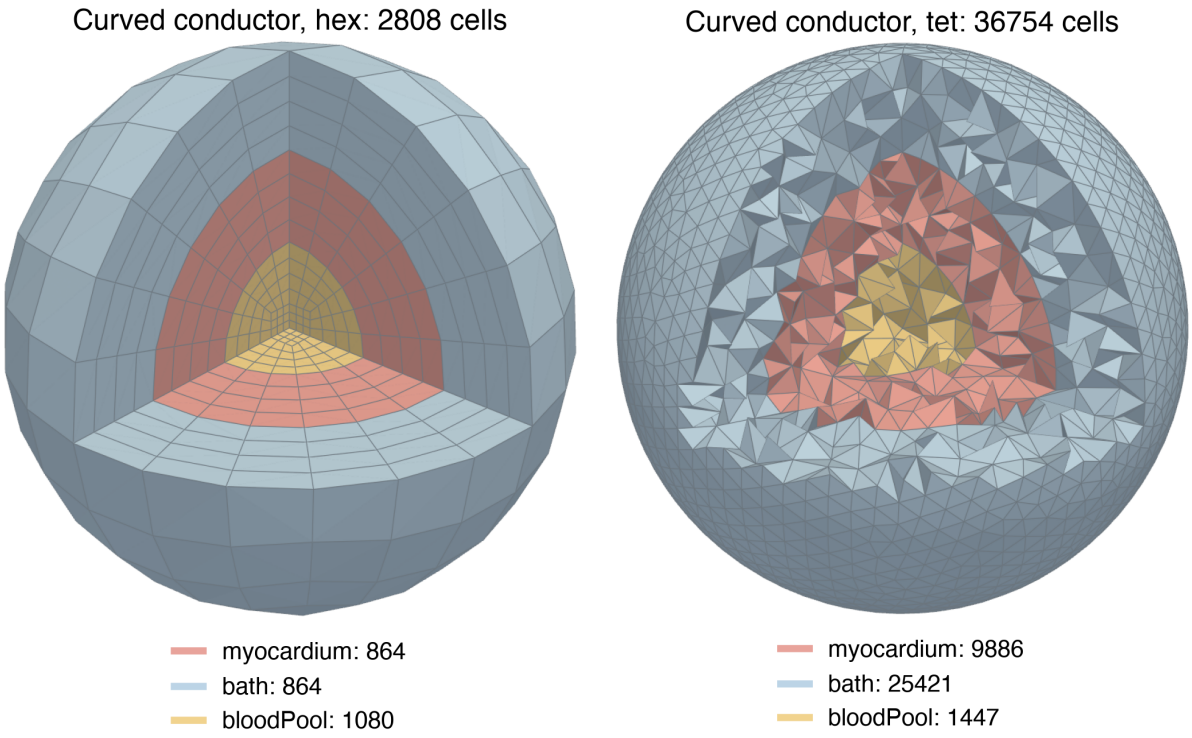


Figure 5: Exact fields on the plane $z = 0$; circles at r_1, r_2, r_3. (a) ϕ at $t = 0$ in all zones, gauged to 0 at the reference point (square, 0.046 above the plane). (b) σ_B in the bath ball and the outer bath. (c) V_m at $t = 0$ in the heart, the input of the lead-field channel.



(a) hex: `-mesh hex -N 8`

(b) tet: `-mesh tet -N 8`

Figure 6: Meshes of the tutorial `leadFieldECG/sphericalShell` at $N = 8$ (the tutorial default is $N = 12$), coloured by cellZone, made by its `Allrun`: (a) the `blockMesh` cubed sphere, an inner cube and three six-block layers with the faces at r_1, r_2, r_3 projected onto the spheres, $3N/4$ cells along a cube-face edge and $N/2$ radially per layer; (b) `gmsh` Delaunay tetrahedra of size $1/N$, seed 1, with both spheres as shared internal faces, converted by `gmshToFoam`. The octant facing the viewer is removed cell by cell.

3.7 Regeneration

```
H=../../../src/verificationModels/bathBidomainVerification
python shell_mms.py --check $H/manufacturedShellTransmissionBathBidomainReference.H
python -m pytest test_shell_mms.py
python docs/leadField/leadfield_reference.py --check-run SHELL_RUN
python docs/leadField/figures.py --mesh-case shell_hex=SHELL_HEX \
    --mesh-case shell_tet=SHELL_TET
```

`--write PATH` rewrites the whole header, including a table of self-test samples that the cardiacFoam verifiers compare with their compiled functions. The generator emits no NumPy module and no LaTeX fragments; the formulas above are transcribed from `expressions()` of `shell_mms.py`.

4 The lead-field identity and the static rows

4.1 Description and limitations

Both problems compare the lead-field channel with a direct solve of the same discrete conductor, and both with the exact potential. The comparison with the direct solve is algebraic: it holds on any mesh to the tolerance of the fixed points, and it checks the adjoint, not the discretisation. The comparison with the exact potential measures the discretisation of the conductor and the recording at a cell.

4.2 The identity

The finite-volume conductor is a linear system $A\boldsymbol{\phi} = \mathbf{b}(V_m)$ over all cells, with the reference cell pinned to ϕ_{ref} and $\mathbf{b} = -\nabla_h \cdot (\mathbf{G}_i \nabla_h V_m)$ in heart cells and zero in torso cells, both per unit volume. $A = M + C$, with M the implicit symmetric part and C the explicit corrections (cross diffusion, non-orthogonal and interface terms). For electrode e, `buildLeadFields` solves the adjoint problem

$$A^{\mathsf{T}}\mathbf{z}_e = \mathbf{e}_{c(e)}, \qquad \text{as the fixed point} \quad M\mathbf{z}_e = \mathbf{e}_{c(e)} - C^{\mathsf{T}}\mathbf{z}_e,$$

and the channel is

$$\phi_e = \sum_c z_{e,c}\, V_c\, b_c(V_m) + c_e = \phi_{c(e)} \quad \text{of the converged forward solve,}$$

with c_e the constant contribution of ground values, surface currents (the layered problem) and a non-zero reference value. In the curved conductor the manufactured data enter as one more cell source,

$$s_c = S_c - \frac{1}{V_c} \sum_{f \in \partial c \cap \partial\Omega} g_{\mathrm{out},f} |\mathbf{S}_f|, \qquad s \leftarrow s - \frac{\sum_c V_c s_c}{\sum_c V_c},$$

with S_c the zone source S_H or S_T at the cell centre and the outer-face current moved into the owner cell; its volume mean is removed so that the pure-Neumann problem is consistent. The direct solve adds s to $\mathbf{b}$ and the channel adds $\sum_c z_{e,c} V_c s_c$.

4.3 What is compared

At each sample the verifier writes, per electrode (the body-surface electrodes and the interior probes alike):

column	meaning
`leadField`	the channel ϕ_e on the computed V_m
`direct`	the forward solve of the same conductor on the computed V_m, assembly and solve repeated until the relative change is below `leadFieldFixedPoint tolerance`, in cell $c(e)$
`exactPoint`	$\phi(\mathbf{r}_e, t)$, gauged at $\mathbf{x}_{\mathrm{ref}}$

column	meaning
`exactCentroid`	$\phi(\mathbf{x}_{c(e)}, t)$, gauged at $\mathbf{x}_{\mathrm{ref}}$
`leadFieldStatic`	the channel on the exact V_m at the heart cell centres (plus $\mathbf{s}$ in the curved conductor)
`directStatic`	the direct solve on the same exact V_m (and $\mathbf{s}$)

and the summary rows (L1, L2 and maximum over the samples, per electrode and as the maximum over the electrodes):

- `leadField-direct` and `leadField-direct_relative` (divided by the largest |`direct`|): the reciprocity gap;
- `leadFieldStatic-directStatic`: the same gap on the exact V_m;
- `leadFieldStatic-exactPoint`, `leadFieldStatic-exactCentroid`, `directStatic-exactPoint`, `directStatic-exactCentroid`: the error against the exact potential at the electrode point and at the electrode-cell centroid, in the gauge of the conductor's reference point;
- `localisation` = `exactPoint` − `exactCentroid`: the part of the point error that comes from recording a cell value at a point inside the cell. It is zero in the slab (constant torso potential) and first order in the cell size otherwise;
- `phiStatic`, `phiStatic_heart`, `phiStatic_torso`, `phiStatic_interfaceBand`, `phiStatic_surface`: mean, volume-weighted RMS and maximum of `directStatic` against the exact potential at every cell centre, over all cells, the heart, the torso, the cells with a heart–torso face and the cells with a face on the outer boundary of the conductor. The last region holds the cells of the body-surface electrodes, so `phiStatic_surface` is the error of the direct solve over the whole body surface, of which the electrodes sample eight cells.

The static rows use the exact V_m, so their error is that of the passive conductor and of the recording alone, not of the myocardium solve. The verifier treats every electrode alike; the summarisers of the studies (`setup/studies/summarize.py` of each case) report the body-surface electrodes, the interior probes and `phiStatic_surface` as separate groups.

4.4 Checks

`buildLeadFields` stops with a fatal error when the probed explicit operator is not linear or does not reproduce the assembled one (10^{-11}), when $\langle \mathbf{z}, C\boldsymbol{\psi} \rangle \neq \langle C^{\mathsf{T}}\mathbf{z}, \boldsymbol{\psi} \rangle$ or M is not symmetric (10^{-12}), when the residual of a lead field exceeds 10^{-8}, or when the pinned row is not satisfied (10^{-8}); it refuses nonlinear schemes. `leadFieldECG` stops when the signature of the lead fields (mesh, conductivities, schemes, electrodes, reference) differs from the case.

References

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